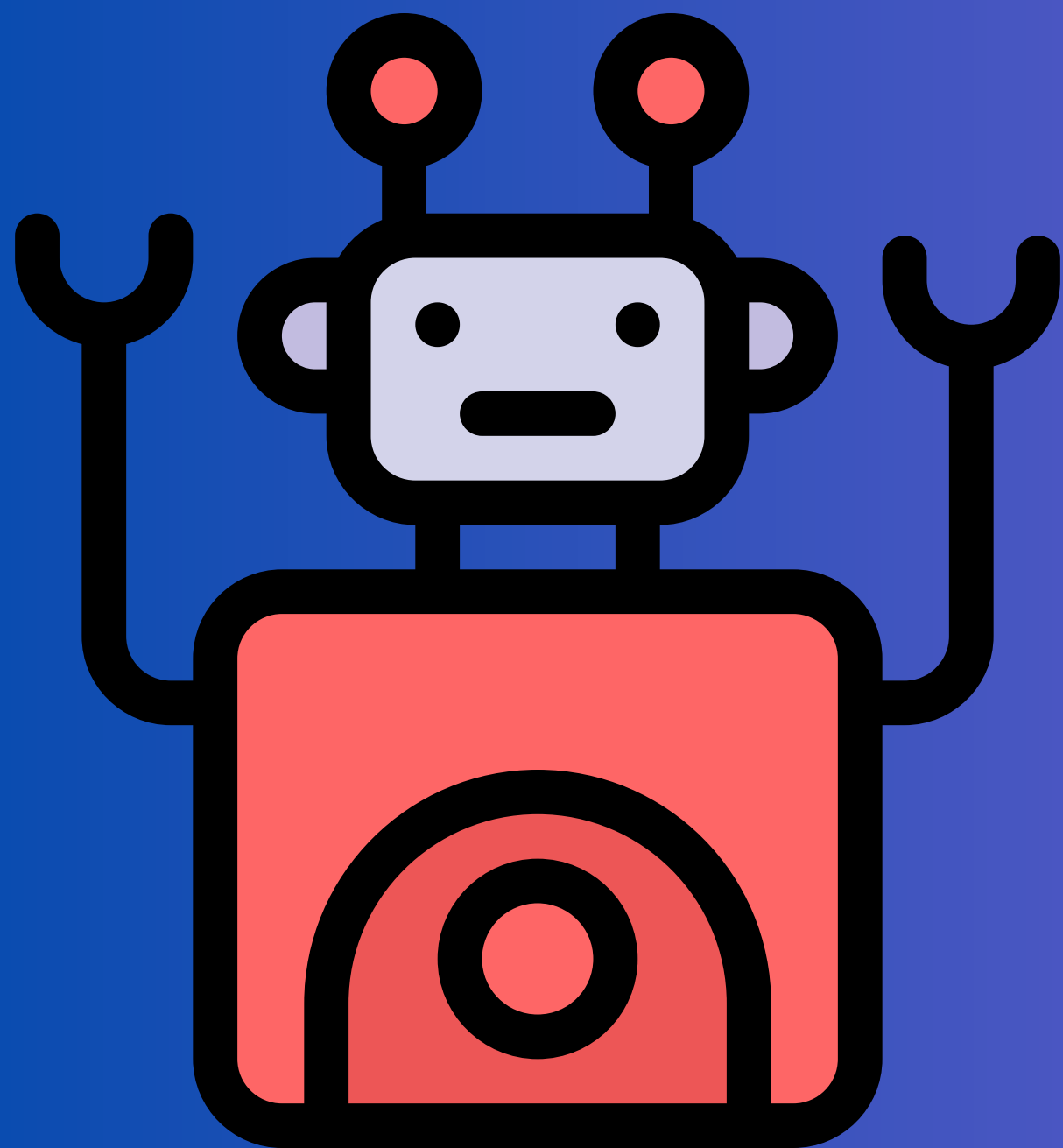




COMPUTER SCIENCE FIELDS IN-DEPTH

WHAT IS GAME DEVELOPMENT?

GAME DEVELOPMENT ASSUMPTIONS

- GAME DEVELOPMENT IS ALL ABOUT CODING
- I NEED TO KNOW COMPLEX MATH FOR GAME DEV
- MAKING GAMES IS FUN ALL THE TIME!



GAME DEVELOPMENT TRUTHS

- WHILE PROGRAMMING IS CRUCIAL, GAME DEV ALSO INVOLVES ART, STORYTELLING, PHYSICS, AND GAME DESIGN PRINCIPLES.
- SOME AREAS (E.G., GRAPHICS, PHYSICS ENGINES) REQUIRE LINEAR ALGEBRA AND CALCULUS, BUT MANY GAMES CAN BE BUILT WITH MINIMAL MATH KNOWLEDGE.
- GAME DEVELOPMENT IS OFTEN TEDIOUS, INVOLVING DEBUGGING, OPTIMIZATION, AND LONG HOURS OF POLISHING SMALL DETAILS.

GAME DEVELOPMENT IS ...

A SPECIALIZED FIELD IN COMPUTER SCIENCE THAT
FOCUSES ON DESIGNING, DEVELOPING, AND DEPLOYING
VIDEO GAMES. IT COMBINES PROGRAMMING, GRAPHICS
RENDERING, ARTIFICIAL INTELLIGENCE, PHYSICS
SIMULATION, NETWORKING, AND USER EXPERIENCE
DESIGN TO CREATE INTERACTIVE DIGITAL EXPERIENCES

GAME DEVELOPMENT TREE

INDIE GAME DEVELOPMENT

AAA GAME
DEVELOPMENT

MOBILE GAME
DEVELOPMENT

VR/AR GAME
DEVELOPMENT

SERIOUS GAMES



POPULAR GAME DEVELOPMENT COMPANIES

- EPIC GAMES
- UNITY TECHNOLOGIES
- NINTENDO
- UBISOFT
- ELECTRONIC ARTS
- ACTIVISION BLIZZARD
- REMEDY ENTERTAINMENT

WHAT IS SOFTWARE ENGINEERING?

SOFTWARE ENGINEERING ASSUMPTIONS

- SWES JUST WRITE CODE ALL DAY
- KNOWING MULTIPLE PROGRAMMING LANGUAGES IS NECESSARY TO BE A GOOD SWE
- IF I CAN CODE WELL, I'LL BE A GREAT SOFTWARE ENGINEER



SOFTWARE ENGINEERING TRUTHS

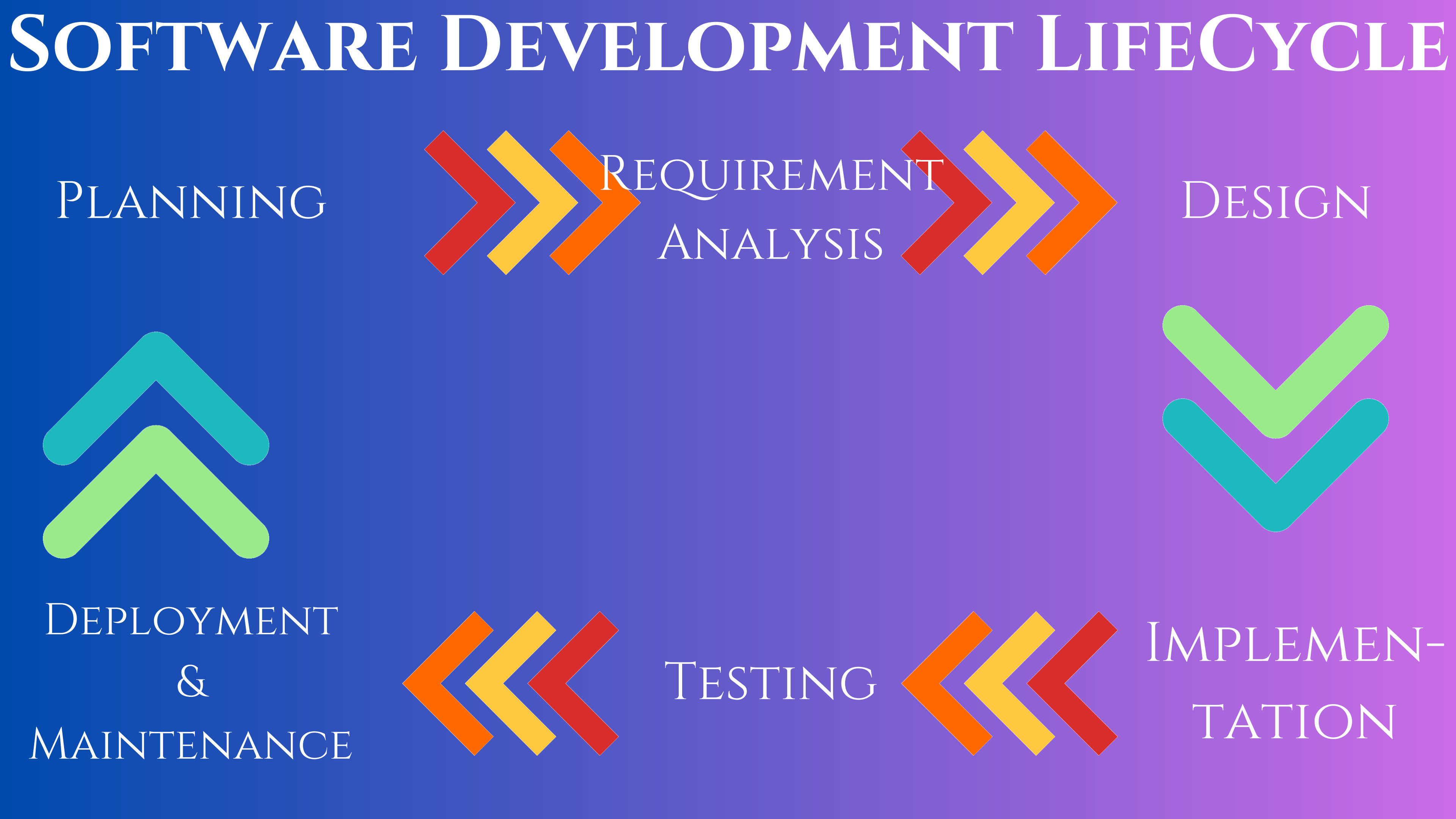
- A LOT OF TIME IS SPENT ON PLANNING, DEBUGGING, TESTING, AND COLLABORATING WITH TEAMS.
- DEEP UNDERSTANDING OF PROBLEM-SOLVING AND SOFTWARE DESIGN IS MORE VALUABLE THAN JUST KNOWING MANY LANGUAGES.
- COMMUNICATION, TEAMWORK, AND UNDERSTANDING USER NEEDS ARE JUST AS IMPORTANT.

SOFTWARE ENGINEERING IS ...

A BRANCH THAT FOCUSES ON DESIGNING, DEVELOPING, TESTING, DEPLOYING, AND MAINTAINING SOFTWARE SYSTEMS IN A STRUCTURED AND EFFICIENT WAY. IT APPLIES ENGINEERING PRINCIPLES TO SOFTWARE DEVELOPMENT, ENSURING RELIABILITY, SCALABILITY, AND MAINTAINABILITY

CORE ASPECTS OF SWE

1. SOFTWARE DEVELOPMENT LIFE CYCLE
2. PROGRAMMING & SOFTWARE ARCHITECTURE
3. VERSION CONTROL & COLLABORATION
4. SOFTWARE TESTING & QUALITY ASSURANCE(QA)
5. AGILE & DEVOPS PRACTICES
6. DATABASE MANAGEMENT & DATA ENGINEERING
7. CLOUD COMPUTING & WEB SERVICES
8. SECURITY & CYBERSECURITY ENGINEERING



SDLC MODELS

- WATERFALL
- V-SHAPED
- ITERATIVE
- SPIRAL
- BIG BANG
- AGILE

FAANG

- FACEBOOK
- AMAZON
- APPLE
- NETFLIX
- GOOGLE

WHAT IS WEB DEVELOPMENT?

WEB DEVELOPMENT ASSUMPTIONS

- WEB DEVELOPMENT IS EASY—JUST HTML AND CSS.
- FRONT-END DEVELOPMENT IS JUST ABOUT MAKING WEBSITES LOOK GOOD.
- I NEED TO LEARN BACKEND AND FRONTEND TO BE A WEB DEVELOPER.



WEB DEVELOPMENT TRUTHS

- MODERN WEB DEV INVOLVES FRAMEWORKS(REACT, ANGULAR), DATABASES, APIS, SECURITY, AND PERFORMANCE OPTIMIZATION.
- IT ALSO INCLUDES UI/UX, ACCESSIBILITY, AND PERFORMANCE OPTIMIZATION.
- YOU CAN SPECIALIZE IN JUST FRONTEND OR BACKEND, THOUGH FULL-STACK DEVELOPERS WORK ON BOTH.

FRONTEND

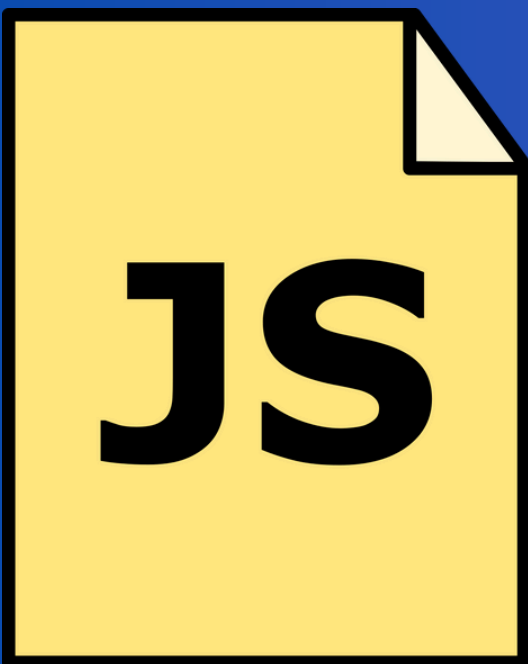
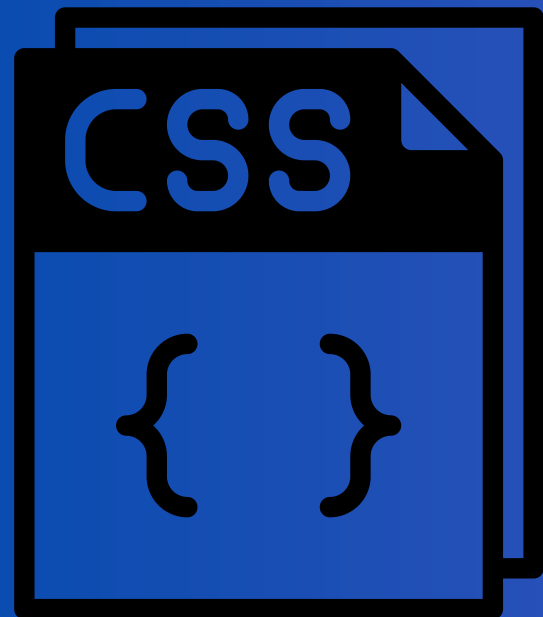


• CLIENT-SIDE LANGUAGES

- HTML
- CSS
- JAVASCRIPT

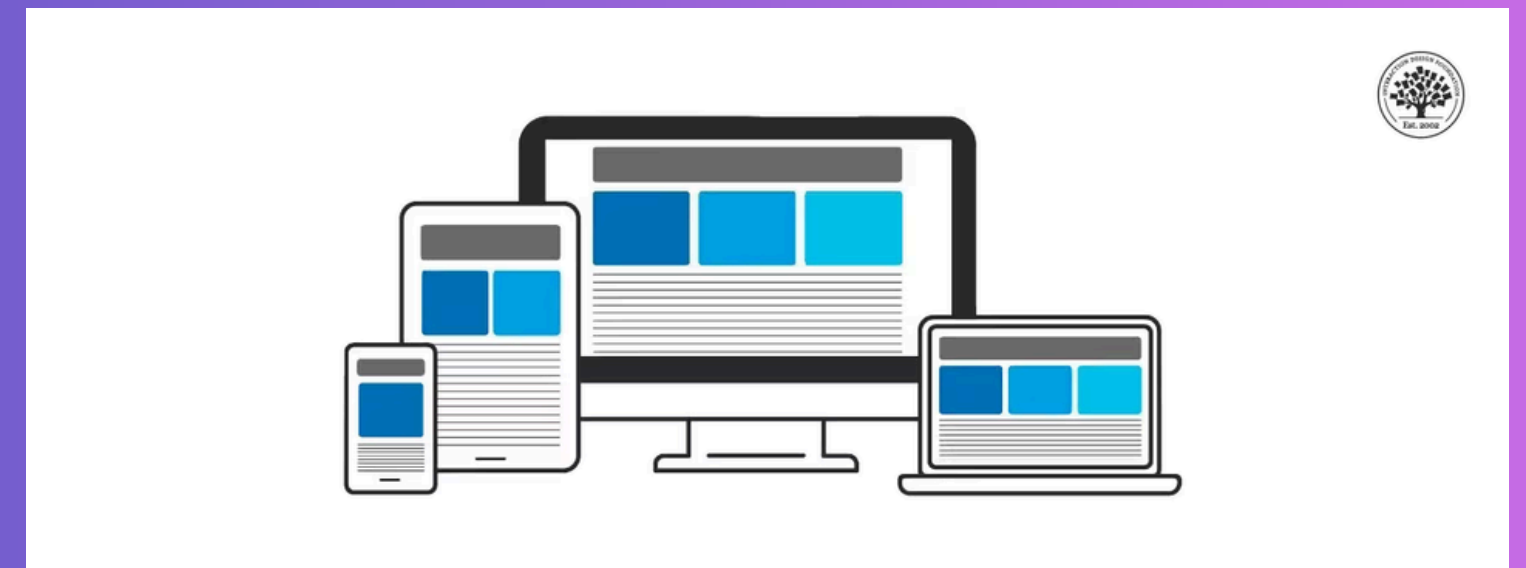
FRAMEWORKS

- REACT.JS
- VUE.JS
- ANGULAR



RESPONSIVE WEB DESIGN

- MOBILE-FIRST
APPROACH
- MEDIA QUERIES &
FLEXIBLE LAYOUTS



BACKEND



• SERVER-SIDE

LANGUAGES

- NODE.JS
- PYTHON
- PHP
- JAVA



APIs

- REST
- GRAPHQL



AUTHENTICATION & SECURITY

- JSON WEB TOKENS
- OAUTH
- PASSPORT.JS
- HTTPS & SECURE HEADERS



DATABASES

- SERVER-SIDE DATABASES
 - SQL
 - MYSQL, POSTGRESQL, SQLITE
 - NOSQL
 - MONGODB
 - FIREBASE
 - ORM
 - SEQUELIZE
 - PRISMA

VERSION CONTROL & COLLABORATION

- GIT
 - TRACKS CHANGES IN CODE
- GITHUB/GITLAB/BITBUCKET
 - CLOUD-BASED REPOSITORIES FOR COLLABORATION
- BRANCHING STRATEGIES
 - FEATURE BRANCHING, GIT FLOW
- PULL REQUESTS & CODE REVIEWS
 - COLLABORATION BEST PRACTICES

MODERN WEB DEV TRENDS

1. RESPONSIVE & MOBILE-FIRST DEVELOPMENT
2. WEB PERFORMANCE OPTIMIZATION
3. PROGRESSIVE WEB APPS(PWAs)
4. SERVERLESS ARCHITECTURE & CLOUD DEPLOYMENT
5. WEB SECURITY ESSENTIALS
6. WEB 3.0 & DECENTRALIZED TECHNOLOGIES
7. AI & MACHINE LEARNING IN WEB DEVELOPMENT
8. LOW-CODE & NO-CODE DEVELOPMENT
9. AR/VR IN WEB DEVELOPMENT

WHAT IS CYBERSECURITY?

CYBERSECURITY ASSUMPTIONS

- CYBERSECURITY IS ALL ABOUT HACKING.
- GOOD CYBERSECURITY JUST MEANS USING STRONG PASSWORDS AND ANTIVIRUS SOFTWARE.
- I NEED TO BE AN EXPERT IN CODING TO WORK IN CYBERSECURITY.



CYBERSECURITY TRUTHS

- ETHICAL HACKING IS JUST ONE PART- THERE'S ALSO NETWORK SECURITY, CRYPTOGRAPHY, RISK MANAGEMENT, AND COMPLIANCE
- SECURITY INVOLVES LAYERED DEFENSE, INCLUDING FIREWALLS, ENCRYPTION, PENETRATION TESTING, AND SECURITY POLICIES
- WHILE PROGRAMMING HELPS, MANY ROLES FOCUS ON SECURITY POLICIES, RISK ASSESSMENT, AND SYSTEM ADMINISTRATION.

CYBERSECURITY IS ...

THE PRACTICE OF PROTECTING SYSTEMS, NETWORKS,
AND DATA FROM CYBER THREATS. IT INVOLVES
IMPLEMENTING SECURITY MEASURES TO PREVENT
UNAUTHORIZED ACCESS, CYBERATTACKS, AND DATA
BREACHES

IMPORTANCE OF CYBERSECURITY

- PROTECTS SENSITIVE INFORMATION (PERSONAL DATA, FINANCIAL RECORDS, GOVERNMENT SECRETS).
- ENSURES THE CONFIDENTIALITY, INTEGRITY, AND AVAILABILITY (CIA TRIAD) OF DATA.
- PREVENTS CYBERCRIME, INCLUDING IDENTITY THEFT, HACKING, AND FRAUD.
- SUPPORTS NATIONAL SECURITY AND THE GLOBAL ECONOMY.

WHAT IS ARTIFICIAL INTELLIGENCE?

ARTIFICIAL INTELLIGENCE

ASSUMPTIONS

- AI IS JUST ABOUT MAKING ROBOTS THINK LIKE HUMANS.
- AI CAN LEARN AND THINK LIKE A HUMAN BRAIN.
- I NEED A DEEP MATH BACKGROUND TO WORK IN AI.



ARTIFICIAL INTELLIGENCE TRUTHS

- AI COVERS MANY AREAS, INCLUDING MACHINE LEARNING, NATURAL LANGUAGE PROCESSING, COMPUTER VISION, AND EXPERT SYSTEMS.
- AI IS POWERFUL BUT LACKS TRUE REASONING, EMOTIONS, AND CONSCIOUSNESS—IT PROCESSES DATA STATISTICALLY RATHER THAN UNDERSTANDING CONCEPTS LIKE HUMANS DO.
- WHILE CONCEPTS LIKE LINEAR ALGEBRA, PROBABILITY, AND CALCULUS HELP, MANY AI TOOLS (E.G., TENSORFLOW, PYTORCH) ABSTRACT A LOT OF THE COMPLEXITY.

ARTIFICIAL INTELLIGENCE ...

REFERS TO COMPUTER SYSTEMS THAT
CAN SIMULATE HUMAN INTELLIGENCE,
INCLUDING LEARNING, REASONING,
PROBLEM-SOLVING, AND DECISION-
MAKING

AI ETHICS AND CHALLENGES

- **BIAS IN AI AND FAIRNESS**
 - **FACIAL RECOGNITION BIAS**
 - **STUDIES HAVE SHOWN THAT AI-POWERED FACIAL RECOGNITION SYSTEMS PERFORM WORSE ON DARKER-SKINNED INDIVIDUALS DUE TO BIASED DATASETS**
 - **BIASED HIRING ALGORITHMS**
 - **AMAZON'S AI HIRING TOOL WAS FOUND TO FAVOR MALE CANDIDATES OVER FEMALE CANDIDATES BECAUSE IT WAS TRAINED ON HISTORICAL HIRING DATA THAT REFLECTED GENDER BIAS**

AI ETHICS AND CHALLENGES

- **AI AND PRIVACY CONCERNS**
 - **CAMBRIDGE ANALYTICA SCANDAL**
 - **AI-DRIVEN DATA ANALYSIS WAS USED TO MANIPULATE POLITICAL CAMPAIGNS BASED ON FACEBOOK USER DATA.**
 - **ALWAYS-ON AI ASSISTANTS**
 - **VIRTUAL ASSISTANTS (E.G., ALEXA, GOOGLE ASSISTANT) CONSTANTLY LISTENING CAN RAISE PRIVACY CONCERNS**

ACTIVITY 2

- THE GROUP WILL LOOK FOR AN UNEXPECTED CAREER IN EACH FIELD:
 - GAME DEVELOPMENT, SOFTWARE ENGINEERING, WEB DEVELOPMENT, CYBERSECURITY, ARTIFICIAL INTELLIGENCE
- WHILE RESEARCHING, ANSWER THE FOLLOWING QUESTIONS:
 - WHAT ARE THE RESPONSIBILITIES OF THAT CAREER? WHAT IS A RESPONSIBILITY THAT YOU DIDN'T EXPECT TO BELONG TO THAT CAREER AND WHY?
 - WHAT COMPANIES OFFER THAT JOB POSITION? WHAT IS THE COMPANY'S MISSION?
 - WHAT IS THE SALARY FOR EACH RESPECTIVE COMPANY? IS IT SURPRISING? WHY OR WHY NOT?
- SUBMIT EITHER A POWERPOINT OR A WORD DOCUMENT TO THE BB PORTAL